Capstone Checkpoint 3.1 RAGAS evaluation results

Oracle, vector, and hybrid experiment comparison. Generated 2026-09-07 11:36 EDT.

19	3	361	19	8
Included runs	Configurations	Question-runs	Questions	Legacy runs excluded

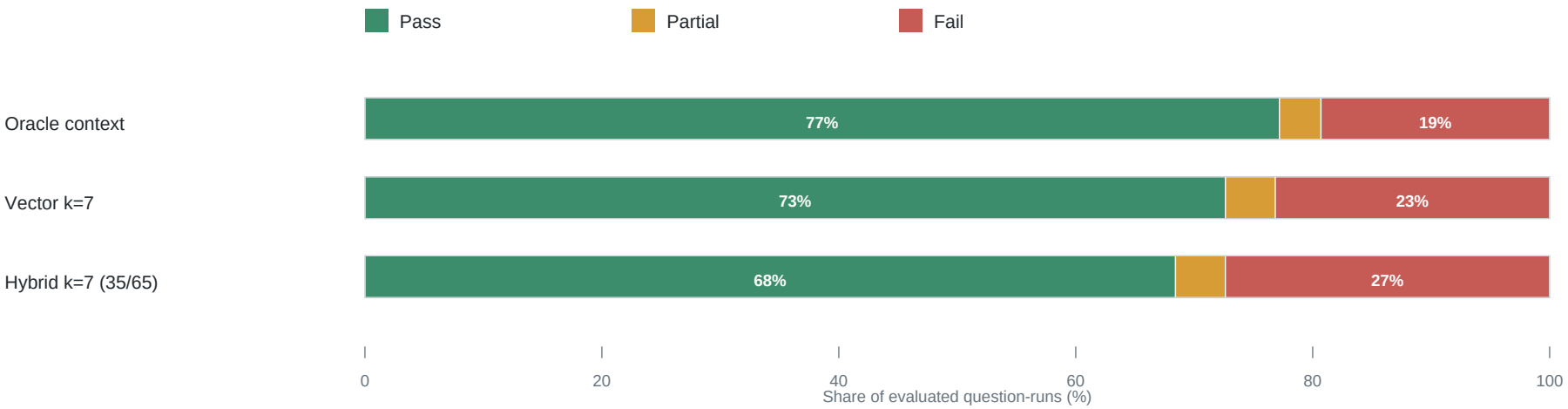
Summary

The highest observed mean quality score was 0.789 for Oracle context across 9 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 89.9% for Vector k=7. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 66.0 for Exhaustive list or time filter under Vector k=7. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

Answer outcomes

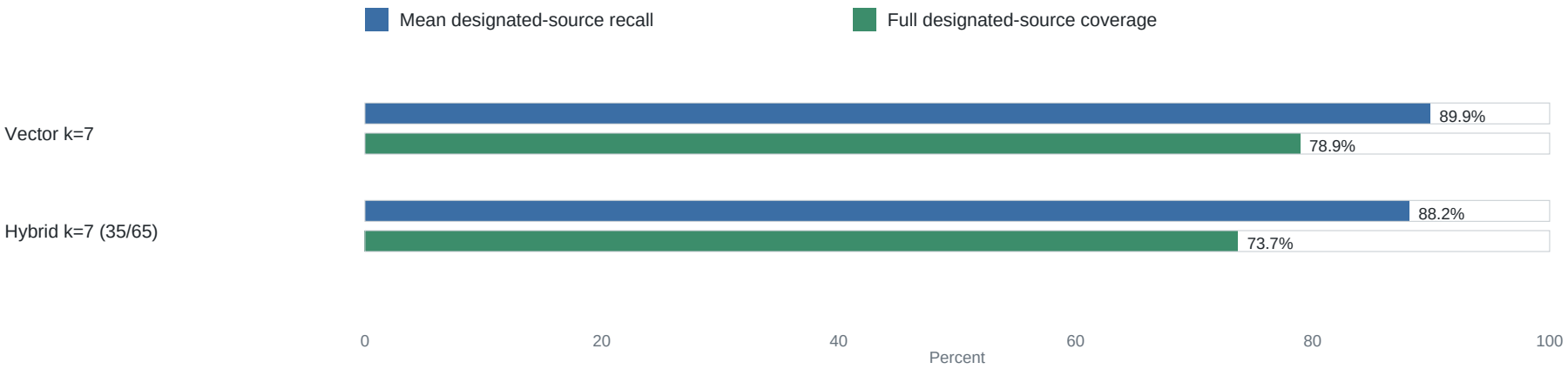


Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
Oracle context	9	171	77.2%	3.5%	19.3%	0.789	0.048	n/a	n/a	63.9k
Vector k=7	5	95	72.6%	4.2%	23.2%	0.747	0.013	89.9%	78.9%	127.0k
Hybrid k=7 (35/65)	5	95	68.4%	4.2%	27.4%	0.705	0.039	88.2%	73.7%	184.6k

Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.

Retrieval diagnostics

Designated-source coverage



The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.

Outcome and source-coverage combinations

Configuration	Pass, full	Pass, incomplete	Non-pass, full	Non-pass, missing
Hybrid k=7 (35/65)	50	15	20	10
Vector k=7	59	10	16	10

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

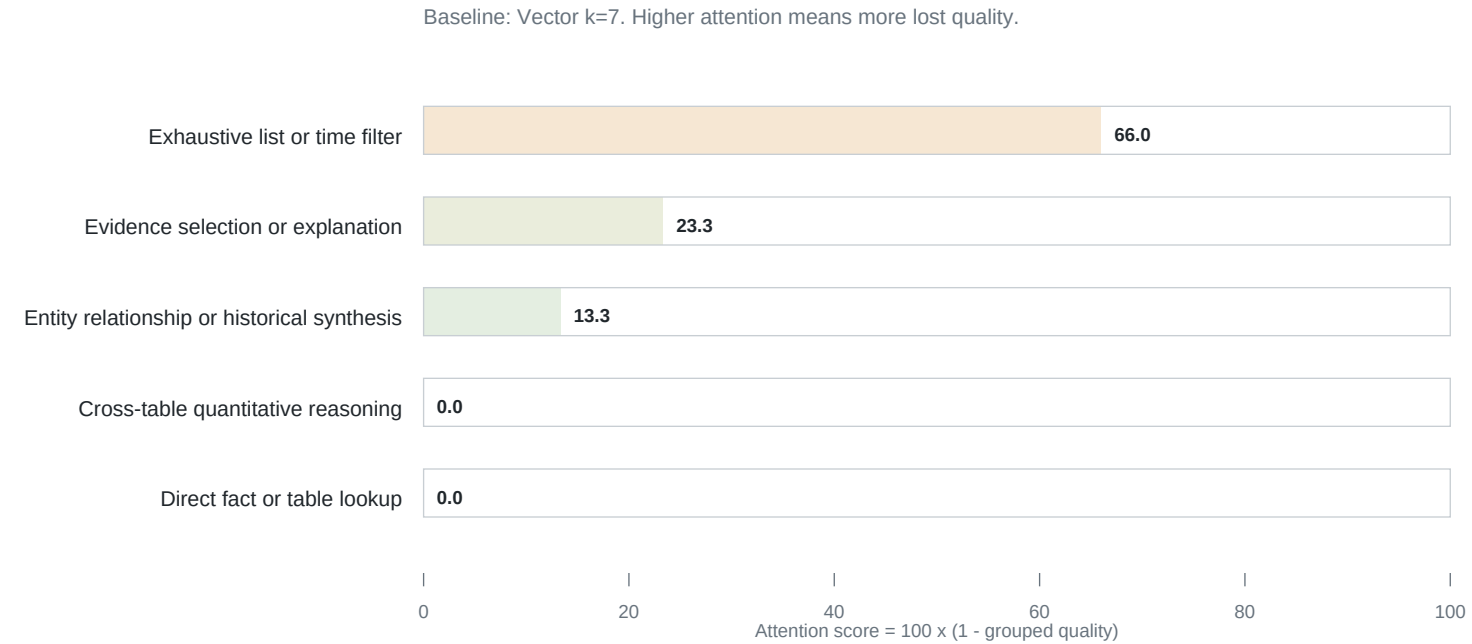
	Oracle context	Vector k=7	Hybrid k=7 (35/65)
GT01	**1.00**	**1.00**	**1.00**
GT02	**0.89**	**1.00**	**0.80**
GT03	**0.00**	**0.00**	**0.00**
GT04	**1.00**	**1.00**	**1.00**
GT05	**1.00**	**1.00**	**1.00**
GT06	**1.00**	**1.00**	**1.00**
GT07	**0.50**	**0.70**	**0.80**
GT08	**0.56**	**0.30**	**0.40**
GT09	**1.00**	**1.00**	**1.00**
GT10	**0.00**	**0.00**	**0.00**
GT11	**0.11**	**0.00**	**0.00**
GT12	**1.00**	**1.00**	**1.00**
GT13	**1.00**	**1.00**	**0.40**
GT14	**1.00**	**1.00**	**1.00**
GT15	**1.00**	**1.00**	**1.00**
GT16	**1.00**	**0.20**	**0.00**
GT17	**1.00**	**1.00**	**1.00**
GT18	**1.00**	**1.00**	**1.00**
GT19	**0.94**	**1.00**	**1.00**

Fail (0) Partial (0.5) Pass (1)

Question	Oracle context	Vector k=7	Hybrid k=7 (35/65)
GT01	1.000	1.000	1.000
GT02	0.889	1.000	0.800
GT03	0.000	0.000	0.000
GT04	1.000	1.000	1.000
GT05	1.000	1.000	1.000
GT06	1.000	1.000	1.000
GT07	0.500	0.700	0.800
GT08	0.556	0.300	0.400
GT09	1.000	1.000	1.000
GT10	0.000	0.000	0.000
GT11	0.111	0.000	0.000
GT12	1.000	1.000	1.000
GT13	1.000	1.000	0.400
GT14	1.000	1.000	1.000
GT15	1.000	1.000	1.000
GT16	1.000	0.200	0.000
GT17	1.000	1.000	1.000
GT18	1.000	1.000	1.000
GT19	0.944	1.000	1.000

Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.



Task family	Questions	Baseline quality	Attention	Fail	Oracle quality	Oracle gap	Article recall
Exhaustive list or time filter	5	0.340	66.0	64.0%	0.322	-0.018	86.7%
Evidence selection or explanation	3	0.767	23.3	20.0%	0.815	+0.048	100.0%
Entity relationship or historical synthesis	6	0.867	13.3	10.0%	1.000	+0.133	87.5%
Cross-table quantitative reasoning	2	1.000	0.0	0.0%	0.972	-0.028	75.0%
Direct fact or table lookup	3	1.000	0.0	0.0%	1.000	+0.000	100.0%

Baseline quality and failure rate use Vector k=7. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.

Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

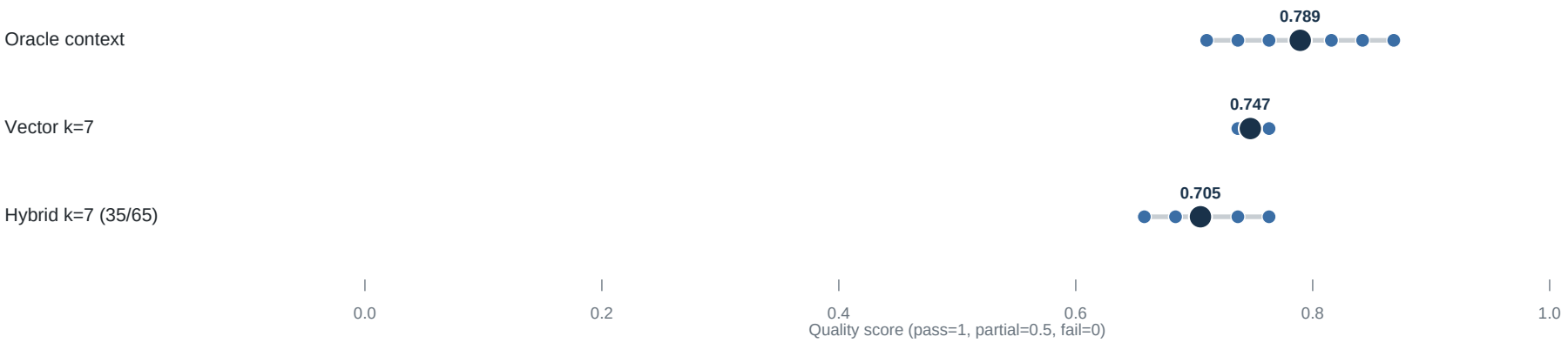
	Oracle context	Vector k=7	Hybrid k=7 (35/65)
1 article	**0.732**	**0.727**	**0.673**
2 articles	**0.989**	**0.840**	**0.800**
3+ articles	**0.667**	**0.667**	**0.667**

Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	Oracle context	11	99	70.7%	5.1%	24.2%	0.732	n/a
2 articles	Oracle context	5	45	97.8%	2.2%	0.0%	0.989	n/a
3+ articles	Oracle context	3	27	66.7%	0.0%	33.3%	0.667	n/a
1 article	Vector k=7	11	55	70.9%	3.6%	25.5%	0.727	100.0%
2 articles	Vector k=7	5	25	80.0%	8.0%	12.0%	0.840	80.0%
3+ articles	Vector k=7	3	15	66.7%	0.0%	33.3%	0.667	69.4%
1 article	Hybrid k=7 (35/65)	11	55	63.6%	7.3%	29.1%	0.673	100.0%
2 articles	Hybrid k=7 (35/65)	5	25	80.0%	0.0%	20.0%	0.800	80.0%
3+ articles	Hybrid k=7 (35/65)	3	15	66.7%	0.0%	33.3%	0.667	58.3%

Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.

Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.



Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
frosty_berners_lee	Oracle context	19	14	1	4	0.763	n/a	n/a
crazy_thiel	Oracle context	19	14	0	5	0.737	n/a	n/a
elated_brin	Oracle context	19	15	1	3	0.816	n/a	n/a
loving_diffie	Oracle context	19	15	0	4	0.789	n/a	n/a
vibrant_ritchie	Oracle context	19	15	1	3	0.816	n/a	n/a
pensive_hopper	Oracle context	19	13	1	5	0.711	n/a	n/a
upbeat_diffie	Oracle context	19	16	1	2	0.868	n/a	n/a
lucid_tarjan	Oracle context	19	14	1	4	0.763	n/a	n/a
amazing_sutherland	Oracle context	19	16	0	3	0.842	n/a	n/a
wonderful_milner	Vector k=7	19	13	2	4	0.737	89.9%	78.9%
confident_valiant	Vector k=7	19	14	0	5	0.737	89.9%	78.9%
sad_bezos	Vector k=7	19	14	0	5	0.737	89.9%	78.9%
jolly_gates	Vector k=7	19	14	1	4	0.763	89.9%	78.9%
eloquent_hellman	Vector k=7	19	14	1	4	0.763	89.9%	78.9%

Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
heuristic_hellman	Hybrid k=7 (35/65)	19	14	1	4	0.763	88.2%	73.7%
boring_diffie	Hybrid k=7 (35/65)	19	13	0	6	0.684	88.2%	73.7%
competent_cerf	Hybrid k=7 (35/65)	19	13	2	4	0.737	88.2%	73.7%
nifty_sutherland	Hybrid k=7 (35/65)	19	12	1	6	0.658	88.2%	73.7%
nostalgic_shannon	Hybrid k=7 (35/65)	19	13	0	6	0.684	88.2%	73.7%

Methodology and limitations

Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

Configuration grouping

Runs are grouped by evaluation mode, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

Included experiment files

Run file
oracle_mode_pass_fail_partial/amazing_sutherland.csv
oracle_mode_pass_fail_partial/crazy_thiel.csv
oracle_mode_pass_fail_partial/elated_brin.csv
oracle_mode_pass_fail_partial/frosty_berners_lee.csv
oracle_mode_pass_fail_partial/loving_diffie.csv
oracle_mode_pass_fail_partial/lucid_tarjan.csv
oracle_mode_pass_fail_partial/pensive_hopper.csv
oracle_mode_pass_fail_partial/upbeat_diffie.csv
oracle_mode_pass_fail_partial/vibrant_ritchie.csv
retriever_active_hybryd/boring_diffie.csv
retriever_active_hybryd/competent_cerf.csv

Run file
retriever_active_hybryd/heuristic_hellman.csv
retriever_active_hybryd/nifty_sutherland.csv
retriever_active_hybryd/nostalgic_shannon.csv
retriever_active_vector_search/confident_valiant.csv
retriever_active_vector_search/eloquent_hellman.csv
retriever_active_vector_search/jolly_gates.csv
retriever_active_vector_search/sad_bezos.csv
retriever_active_vector_search/wonderful_milner.csv

Excluded files

Run file	Reason
oracle_mode_pass_fail_only/adoring_shamir.csv	legacy pass/fail-only scoring
oracle_mode_pass_fail_only/blissful_stallman.csv	legacy pass/fail-only scoring
oracle_mode_pass_fail_only/elastic_hellman.csv	legacy pass/fail-only scoring
oracle_mode_pass_fail_only/gifted_kay.csv	legacy pass/fail-only scoring
oracle_mode_pass_fail_only/happy_musk.csv	legacy pass/fail-only scoring
oracle_mode_pass_fail_only/kind_liskov.csv	legacy pass/fail-only scoring
oracle_mode_pass_fail_only/naughty_shannon.csv	legacy pass/fail-only scoring
oracle_mode_pass_fail_only/quizzical_hamilton.csv	legacy pass/fail-only scoring